

The PlayStation motion controller features an orb at the end which can glow in a full range of colours according to the RGB colour model using LEDs. The coloured light serves as an active marker, the position of which can be tracked along the image plane by the PlayStation Eye.

The uniform spherical shape and known size of the lights also allows the system to simply determine the wands' distances from the PlayStation Eye through the image size, thus enabling the wands' positions to be tracked in three dimensions. Combined with internal motion sensors, such as accelerometers and gyroscopes, this allows for highly accurate real-time motion capture with orientation in three dimensions.

The simplicity of the sphere-based distance calculation allows the controller to operate with minimal processing lag, as opposed to other camera-based control techniques on the PS3.

The PlayStation motion controller will feature buttons on the wand, as well as vibration-based haptic technology. In addition to providing a tracking reference, the wand's orb light can be used to simulate aesthetic effects, such as the muzzle flash of a gun, or the paint on a brush.

Using different orb colours for each controller, up to four wands can be tracked at once with the PlayStation Eye. Demonstrations of the controller have featured activities using a single controller, as well as those in which the user wields two controllers, with one wand in each hand.



Using different orb colours for each PS3 motion controller

4.1.2 Project Natal

While Nintendo and Sony battle it out to figure out whose motion-sensitive controller is better, Microsoft has more ambitious plans with Project Natal – an add-on for the Xbox 360. It sounds like an exaggeration right now and perhaps we are placing our expectations a bit high, but we dare say that this generation of consoles will be remembered more for the Project Natal's motion-sensing technology than the Nintendo Wii and its WiiMote.

It all started with a subtle hit by Microsoft's founder and ex-CEO Bill Gates during a free-wheeling chat at the AllThingsD conference in 2007, when he casually remarked: "Imagine a game machine where you're just going to pick up